

House Price Prediction — Summary of Findings

Internship Project · Week 1 · Dataset: Housing.csv (545 rows, 13 columns)

OBJECTIVE

Build and evaluate machine learning models to predict house prices from property features (area, bedrooms, bathrooms, amenities, etc.), identify the strongest price drivers, and translate the results into actionable insight for a real-estate business.

METHODOLOGY

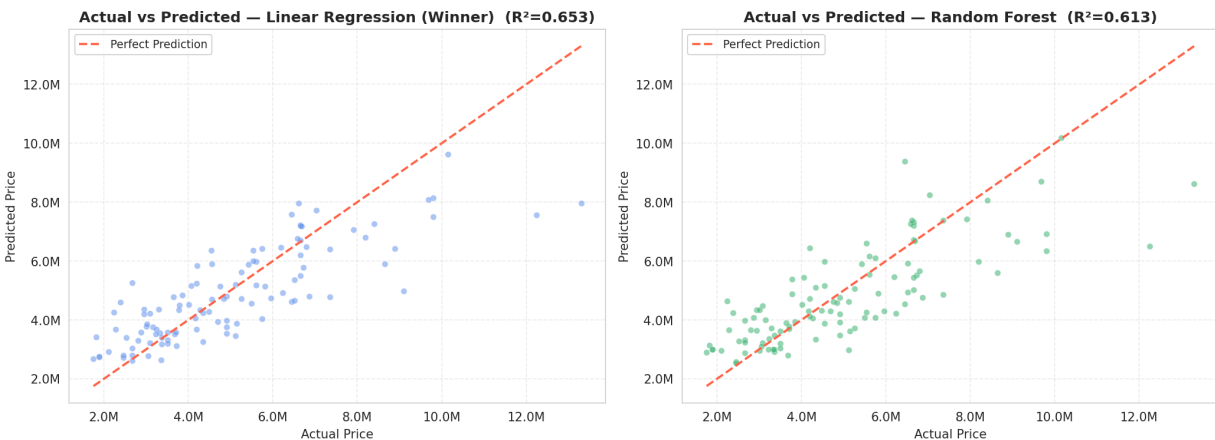
Data was cleaned (missing-value check, duplicate removal, One-Hot Encoding of categorical features) and split 80/20 into training and test sets (random_state=42). Two regression models were trained and compared: **Linear Regression** and **Random Forest Regressor**, evaluated using MAE, RMSE, and R-Squared.

MODEL PERFORMANCE

Model	MAE	RMSE	R-Squared
Linear Regression (Best)	970,043	1,324,507	0.6529
Random Forest	1,013,969	1,398,116	0.6133

Linear Regression was the best-performing model, explaining ~65% of the variance in house prices with the lowest average error — a sign that price in this dataset follows largely linear relationships rather than complex non-linear patterns.

Model Prediction Accuracy Comparison — Winner: Linear Regression



KEY INSIGHTS

- **Area is the dominant price driver**, accounting for ~47% of feature importance — far ahead of any other factor.
- **Bathrooms (15.3%)** and **air conditioning (6.0%)** are secondary but meaningful contributors to price.
- Price distribution is **right-skewed** (mean Rs. 4.77M vs. median Rs. 4.34M) — a smaller number of high-value homes pull the average upward.
- Bedroom count had surprisingly **low influence** on price compared to total area, despite being commonly assumed important by buyers.

BUSINESS RECOMMENDATION

Real-estate pricing and marketing efforts should prioritize **livable area** above all other attributes, since it overwhelmingly drives buyer valuation. Bathrooms and air conditioning offer a secondary lever for premium positioning, while bedroom count and most other amenities have limited standalone impact on price.